Electromagnetic–Acoustic Model of PWM-Excited Differential Forces on a 120 m Single-Core Cable Run

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Abstract

This note extends the cable model of the companion document to the acoustic domain. The audible emission observed near the converter cables is modeled as the end of a four-stage chain: (i) the differential PWM current ripple, (ii) the Ampère forces it produces between the three single-core conductors, (iii) the transverse mechanical vibration of the cable spans, and (iv) the acoustic radiation of the vibrating conductors. The force is shown to be governed by the *differential* (positive/negative-sequence) currents, not the common mode, confirming the field observation. The model reproduces the qualitative evidence: a tonal line at the switching frequency, its conversion to a broadband "flowing-water" sound under random PWM, and the presence of emission on the 120 m section while the 40 m section is silent. The electrical parameters of the companion document are retained unchanged.

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1 Scope and physical hypothesis

The site evidence is the following:

1. the sound tracks the inverter PWM; with random (spread-spectrum) PWM the tonal structure disappears and the emission becomes broadband, perceptually like flowing water;

2. the 120 m cable section emits, the 40 m section does not;
3. a tonal line coincides with the fixed converter switching frequency.

The working hypothesis, examined and confirmed below, is that the excitation is the *differential* PWM current ripple acting through inter-conductor Ampère forces, and that the length dependence arises from the mechanical/acoustic response, not from the electromagnetic source itself.

The arrangement modeled is three *separate* single-core copper cables, 90 mm^2 each, length $\ell = 120$ m. Separate single cores are essential to the mechanism: unlike the bound conductors of a single three-core cable, each single core is mechanically free to vibrate between its supports, so the inter-conductor force does work on a movable structure. The per-unit-length electrical parameters are taken unchanged from the companion document (in particular $C_g = 36$ nF retained as requested, although the common-mode capacitance does not enter the force).

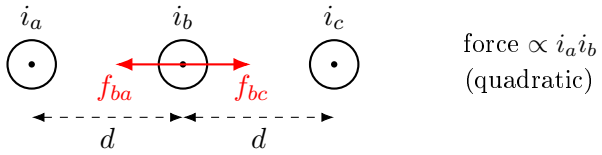


Figure 1: Transverse Ampère forces on a single core from its two neighbors. The force on each conductor is the vector sum of two contributions, each proportional to the product of conductor currents.

2 From currents to forces

Two long parallel conductors j, k at separation d_{jk}, carrying currents i_j, i_k, exert on each other the transverse force per unit length

$$f_{jk}(t) = \frac{\mu_0}{2\pi\, d_{jk}}\, i_j(t)\, i_k(t), \tag{1}$$

attractive for like signs. This is the same magnetostatic law used for short-circuit force calculations (IEC 60865). The net transverse force per unit length on conductor a is the vector sum from its neighbors,

$$\boldsymbol{f}_a(t) = \frac{\mu_0}{2\pi}\left[\frac{i_a i_b}{d_{ab}}\,\hat{\boldsymbol{u}}_{ab} + \frac{i_a i_c}{d_{ac}}\,\hat{\boldsymbol{u}}_{ac}\right], \tag{2}$$

with $\hat{\boldsymbol{u}}$ the unit vectors in the conductor cross-section (Fig. 1). The essential feature is the *quadratic* dependence on current: the force spectrum is the autocorrelation of the current spectrum, so it contains sums and differences of all current frequency components.

3 Force spectrum and the dominance of the differential mode

Write each phase current as a fundamental term plus PWM ripple,

$$i_k(t) = \hat{I}_1 \cos(\omega_1 t - \varphi_k) + \tilde{\imath}_k(t), \qquad \varphi_k \in \{0, \tfrac{2\pi}{3}, \tfrac{4\pi}{3}\}, \tag{3}$$

where $\tilde{\imath}_k$ has its spectral energy concentrated at the carrier f_{sw} and its multiples, with sidebands at $\pm m f_1$ (double-Fourier PWM spectrum). Expanding the product in (1),

$$i_a i_b = \underbrace{\hat{I}_1^2 \cos(\omega_1 t)\cos(\omega_1 t - \tfrac{2\pi}{3})}_{\text{(I) DC+2}f_1} + \underbrace{\hat{I}_1\left[\cos(\omega_1 t)\,\tilde{\imath}_b + \cos(\omega_1 t - \tfrac{2\pi}{3})\,\tilde{\imath}_a\right]}_{\text{(II) fundamental}\times\text{ripple}} + \underbrace{\tilde{\imath}_a\tilde{\imath}_b}_{\text{(III) ripple}^2}. \tag{4}$$

The three groups give three force bands:

(I) a DC offset and a $2f_1$ component (100 Hz) — the classical conductor "hum". It is the largest in amplitude but radiates poorly (see Sec. 6) and lies below the perceptually critical band.

(II) components at $f_{sw} \pm m f_1$, i.e. *centered on the switching frequency*, of amplitude $\propto \hat{I}_1\, \hat{\tilde{i}}_{sw}$. This is the audible tonal excitation.

(III) components at $2f_{sw}$, DC, and difference frequencies, of amplitude $\propto \hat{\tilde{i}}_{sw}^2$, smaller because the ripple is small compared with the fundamental.

The dominant audible force is therefore band (II), whose amplitude per unit length is

$$\boxed{\hat{F}_{sw} \approx \frac{\mu_0}{2\pi d}\, \hat{I}_1\, \hat{\tilde{i}}_{sw}} \tag{5}$$

at frequency $\approx f_{sw}$ with $\pm f_1$ sidebands.

Differential vs. common mode. Equation (5) settles the field intuition. The force is the product of a *large* differential fundamental current $\hat{I}_1$ (hundreds of amperes) and the differential ripple $\hat{\tilde{i}}_{sw}$. The common-mode current is small — it is limited by the high common-mode impedance of the shield/earth loop — and, being squared or multiplied by itself in (4), contributes a force smaller by the ratio $(\hat{I}_{cm}/\hat{I}_1)^2$ or $(\hat{I}_{cm}/\hat{I}_1)$. The audible mechanical excitation is thus governed by the positive/negative-sequence (differential) currents; the common mode matters for the shield/EMC path of the companion document, not for the acoustic force.

4 Coupling to the electrical cable model

The ripple amplitude $\hat{\tilde{i}}_{sw}$ entering (5) is set by the volt-seconds the converter applies across the total per-phase series inductance during a switching subinterval,

$$\hat{\tilde{i}}_{sw} \sim \frac{V_{dc}}{8\, L_\sigma\, f_{sw}}, \qquad L_\sigma = L_{\text{cable}} + L_{\text{load}}, \tag{6}$$

plus any resonant amplification from the cable shunt capacitance. With the companion-document value $L_{\text{cable}} = L_1' \ell = 30$ μH at 120 m (and 10 μH at 40 m), the cable inductance is a minor addition to the motor leakage; the *source* ripple is nearly the same on both sections. The length dependence of the *sound* must therefore originate downstream, in the mechanical and acoustic response (Secs. 5–8), not in the electromagnetic forcing.

A secondary electrical effect is the differential resonance between L_σ and the cable capacitance $C_1 = C_1' \ell$, at $f_0 = 1/(2\pi\sqrt{L_\sigma C_1})$. For the reference values this lies in the tens of kilohertz (well above $f_{sw} = 3.8$ kHz) and is not the primary audible mechanism, but it should be checked against the actual L_σ because it scales as $\ell^{-1/2}$ and moves downward with length.

5 Mechanical model of the cable span

Each single core between two supports (cleats, span L_s) responds to the distributed transverse force $f(z,t)$ as a beam under tension,

$$\rho_A \frac{\partial^2 w}{\partial t^2} + EI \frac{\partial^4 w}{\partial z^4} - T \frac{\partial^2 w}{\partial z^2} = f(z,t), \tag{7}$$

with w the transverse displacement, ρ_A the mass per unit length, EI the (low, because stranded) bending stiffness and T the mechanical tension. The undamped natural frequencies are, in the two limits,

$$\text{string: } f_n = \frac{n}{2L_s}\sqrt{\frac{T}{\rho_A}}, \qquad \text{beam: } f_n = \frac{(\beta_n L_s)^2}{2\pi L_s^2}\sqrt{\frac{EI}{\rho_A}}, \tag{8}$$

with $\beta_n L_s = \{4.73, 7.85, \dots\}$ for clamped ends. Resonant amplification of the displacement occurs when a force harmonic $f_{sw} \pm m f_1$ coincides with an f_n; the modal velocity at resonance is

$$\hat{v}_{\mathrm{res}} \approx \frac{\hat{F}_{sw}}{\rho_A\, \omega_{sw}}\, Q, \tag{9}$$

Q being the mechanical quality factor of the span ($Q \sim 10$–50 for a lightly damped cleated conductor). Off resonance the same expression holds with $Q \to 1$.

Modal density. For a one-dimensional waveguide of total length ℓ, the number of modes per unit frequency scales with length,

$$\frac{\mathrm{d}N}{\mathrm{d}f} \propto \ell \qquad \Longrightarrow \qquad N_{120} \approx 3\, N_{40}. \tag{10}$$

This is the key length-dependent quantity (Sec. 8).

6 Acoustic radiation

A conductor of radius a vibrating transversely is, acoustically, a line of dipoles. The radiated sound power from a length L is

$$W_{ac} = \rho_0 c\, \sigma_{\mathrm{rad}}\, S\, \overline{v^2}, \qquad S = 2\pi a L, \qquad \overline{v^2} = \tfrac{1}{2}\hat{v}^2, \tag{11}$$

with $\rho_0 c = 415$ Pa s/m and σ_{rad} the radiation efficiency. For transverse (dipole) motion σ_{rad} rises with the compactness ratio $ka = 2\pi f a/c$. At $f_{sw} = 3.8$ kHz and $a = 9$ mm, $ka \approx 0.63$, giving an order-of-unity, moderately efficient radiator ($\sigma_{\mathrm{rad}} \sim 0.2$); the 100 Hz hum, by contrast, has $ka \approx 0.017$ and is an extremely poor radiator, which is why the audible emission is the switching tone and not the (larger) 100 Hz force. The sound power level is $L_W = 10 \log_{10}(W_{ac}/10^{-12})$ dB.

7 Numerical estimate

Representative site values (to be replaced by measured data) are listed in Table 1.

Table 1: Representative parameters for the estimate.

Quantity	Symbol	Value
Fundamental current (peak)	$\hat{I}_1$	283 A (200 A rms)
Carrier-harmonic ripple (amp.)	$\hat{\tilde{i}}_{sw}$	20 A
Conductor separation	d	40 mm
Fundamental / switching freq.	f_1, f_{sw}	50 Hz, 3.8 kHz
Cable mass per length	ρ_A	1.3 kg/m
Conductor outer radius	a	9 mm
Mechanical quality factor	Q	1 (off-res.) / 30 (res.)
Radiation efficiency	σ_{rad}	0.2

Forces. From (5) and the $2f_1$ part of (4), with $\mu_0/(2\pi d) = 5 \times 10^{-6}$ N/A^2:

$$\hat{F}_{2f_1} = \frac{\mu_0}{2\pi d}\frac{\hat{I}_1^2}{2} = 0.20 \text{ N/m} \quad (100 \text{ Hz}), \qquad \hat{F}_{sw} = \frac{\mu_0}{2\pi d}\hat{I}_1 \hat{\tilde{i}}_{sw} = 0.028 \text{ N/m} \quad (\approx f_{sw}).$$

Velocity. From (9), $\hat{v} = 0.9$ μm/s off resonance ($Q = 1$) and 27 μm/s at resonance ($Q = 30$).

Sound. From (11), Table 2 compares the two sections. The model predicts an audible tone ($\sim$ 42 dB at 1 m) on the 120 m section when a span mode is excited, and an inaudible level ($\sim$ 8 dB) on the non-resonant 40 m section.

Table 2: Predicted radiated levels (SPL at 1 m, point-source approximation).

Section	State	$\hat{v}$	L_W	SPL@1 m
120 m	resonant (Q=30)	27 μm/s	53 dB	$\approx$ 42 dB
40 m	off-resonance (Q=1)	0.9 μm/s	19 dB	$\approx$ 8 dB
40 m	(if resonant)	27 μm/s	49 dB	$\approx$ 38 dB

8 Explanation of the field observations

Why 120 m emits and 40 m is silent. The per-unit-length force is the same on both sections (same currents, separation and carrier). Three length-dependent factors act downstream:

1. **Modal coincidence (dominant).** By (10) the long run has three times the modal density. Under *fixed* (narrow-band) PWM the tonal force is far more likely to coincide with a lightly damped span mode of the 120 m run; the sparser modes of the 40 m run can fall in a gap, leaving it near the off-resonance level. The resonant/non-resonant contrast in Table 2 is a 34 dB difference.
2. **Radiating area.** For equal surface velocity the longer cable radiates $10\log_{10}(120/40) \approx$ 4.8 dB more.
3. **Installation/damping.** Longer free spans or lower clamping damping on the 120 m run lower f_n and raise Q, both increasing emission.

Why random PWM gives a "water" sound. Random/spread-spectrum PWM replaces the discrete carrier line with a broadband spectrum. By (4), band (II) of the force is correspondingly spread across frequency, so instead of driving one resonant mode it distributes energy over many modes at low spectral density. The radiated field loses its tonal character and becomes broadband noise — perceptually the observed flowing-water sound — and, crucially, the peak modal excitation drops, which is the same mechanism by which random PWM is used to reduce acoustic noise in drives.

9 Confirmation and mitigation

To confirm the model.

- Measure the acoustic spectrum and check the tonal line moves with f_{sw} and its sidebands sit at $f_{sw} \pm m f_1$.
- A simple bump/impact test on a 120 m span identifies the span natural frequencies; coincidence of one f_n with $f_{sw} \pm m f_1$ confirms the mechanism.
- Verify the level scales with $\hat{I}_1\,\hat{\tilde{\imath}}_{sw}$ (load and f_{sw} dependence) as predicted by (5).

Mitigation levers (from the model).

- Move f_{sw} off the coincident span mode, or adopt random/spread-spectrum PWM (spreads band (II); lowers peak excitation).
- Detune the mechanical resonance: change cleat spacing L_s (shifts f_n in (8)) or add damped clamps (raises losses, lowers Q).
- Reduce the force: increase conductor separation d ((5) scales as $1/d$), or reduce $\hat{\tilde{\imath}}_{sw}$ with a higher f_{sw} or an output dv/dt/sine filter.

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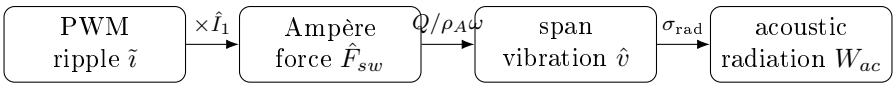
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A Signal chain



The four transfer factors are: the differential fundamental current $\hat{I}_1$ (force generation, Eq. (5)); the mechanical mobility $Q/(\rho_A\omega)$ (Eq. (9)); and the radiation efficiency σ_{rad} (Eq. (11)). The length ℓ enters through the modal density (10) (probability of resonance) and the radiating area (11).